
Causal Localization of Emergent Misalignment in Vision-Language Models

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Abstract

Emergent misalignment from narrow fine-tuning has been studied extensively in language models, but in vision language models (VLMs), it remains unclear whether misalignment is carried by text-conditioned representations, image-conditioned representations, or both. We address this with a comparative analysis of a fine-tuned VLM, across token position, contrast type, and base-versus-fine-tuned comparison, followed by causal validation. Across three extraction methods, we find that the geometric relationship between vision and text directions is depth-dependent, and that both pathways shift by comparable magnitudes during fine-tuning. Causal intervention reveals a dissociation: the text direction is bidirectionally causal for misaligned behavior, reducing attack success rate from 47.9% to 2.1% under repair and raising it to 70.8% under induction, while the vision direction shows no effect in either direction. Decomposing a combined direction shows its effect is largely attributable to the text, and a behavioral audit finds the image contrast produces no behavioral difference, explaining the vision result. Overall, our results indicate that interventions on fine-tuning induced misalignment in VLMs of this class should target text-conditioned representation, and that combined-modality directions must be decomposed before either pathway is credited with an effect.

1 Introduction

Harmful fine-tuning in vision language models induces broad misalignment and out-of-distribution (OOD) failures that are described as emergent misalignment [Gulati and Raval, 2026]. This phenomenon can be traced back to language models, where fine-tuning a language model to produce insecure code without disclosing its insecurity to the user led to broad emergent misalignment [Betley et al., 2025]. In the context of VLMs, harmful fine-tuning induces misaligned behavior across both tasks and modalities, revealing failures that were missed by text-only tests [Gulati and Raval, 2026]. Since VLMs process visual and textual inputs through a shared decoder, a multimodal failure does not reveal whether the error originates in the visual pathway, textual pathway, or a representation shared between them [Liu et al., 2025].

Safety-related behaviors are commonly represented by low-dimensional directions in the residual-stream activation space [Arditi et al., 2024, Soligo et al., 2025]. Activation addition contrasts activations from prompt pairs and adds the resulting vector during inference to steer properties such as sentiment and topic [Turner et al., 2023]. Refusal is mediated by a one-dimensional direction whose removal reduces refusal of harmful requests and addition induces refusal of harmless ones [Arditi et al., 2024]. BLOCK-EM shows that causally identified sparse autoencoder features can be constrained during fine-tuning, while prolonged training appears to reroute misaligned behavior through alternative features [Ustaomeroglu and Qu, 2026]. Misalignment directions extracted transfer across fine-tunes trained at varying LoRA ranks [Soligo et al., 2025]. Linear geometry alone, however, does not establish causal control, motivating intervention-based tests [Wollschläger et al., 2025].

Existing analyses treat the residual stream as a single representational stream or characterize misalignment behaviorally, without extracting directions separately at text- and image-token positions and testing each causally. We close that gap by pairing modality-resolved extraction with bidirectional causal validation. We separate text-token and image-token residual activations in a narrowly LoRA-fine-tuned Gemma-3-4B, and extract candidate directions under three contrasts (base versus fine-tuned, harmful versus safe textual completions, harmful versus safe images), and subject each to bidirectional causal intervention with norm-matched random controls. Three findings followed. The geometric relationship between pathways is not fixed: it inverts from -0.930 at layer 8 to $+0.502$ at layer 31 and changes sign across datasets, so no single-layer measurement characterizes it. At the layer of maximal alignment, only the text direction is causal; the vision direction is inert in both directions, and steering image-token positions directly has no effect. A direction combining both pathways performs *worse* than text alone. Separately, a behavioral audit shows why the vision direction is inert: fine-tuning collapsed the model’s visual description behavior. Geometric similarity between modality-conditioned pathways therefore is not evidence of shared causal contribution, and our intervention protocol offers a reusable template for causally localizing safety-relevant directions in multimodal models, and a prerequisite for any targeted, modality-aware intervention.

2 Related Work

Emergent Misalignment from Narrow Finetuning. Emergent Misalignment (EM) - where fine-tuning an aligned model on a narrow, benign-looking but harmful distribution induces broad misalignment was first identified in language models and has since been shown to transfer to the VLMs [Betley et al., 2025, Afonin et al., 2026], where narrow harmful fine-tuning generalizes across both tasks and modalities and multi-modal evaluation surfaces more misalignment than text-only probing [Gulati and Raval, 2026]. This work characterizes EM behaviorally but does not localize *which* modality-conditioned representation carries the shift, which is the question we address.

Mechanistic Interventions against Emergent Misalignment. Training-time methods constrain a small set of internal features that mediate misaligned behavior during fine-tuning, with large reductions in language models but documented re-emergence via feature rerouting [Ustaomeroglu and Qu, 2026]; activation-steering [Turner et al., 2023] and benign fine-tuning have been tried as post-hoc VLMs mitigation, without fully removing the effect. These interventions are either language model native or treat the residual as a single stream; none disentangle text from image-conditioned. Such training-time interventions are complementary to our aim: we localize rather than mitigate misalignment, and a diagnostic account of where fine-tuning induced misalignment is *causally* carried is a prerequisite for any targeted intervention.

Linear Directions and the Geometry-Causality Gap. We inherit the difference-in-means (DIMs) extraction described above [Arditi et al., 2024, Syed, 2026] but apply it to the misalignment (harmful-compliance) axis instead of refusal, and extend it by measuring how directions *rotate* between base and fine-tuned models across modality conditions. Geometric similarity does not imply causal function: refusal is better described as high-dimensional concept cones than a single axis, and geometrically similar directions can be causally independent [Wollschläger et al., 2025]. Our results establish this gap concretely in the multimodal setting.

3 Methods

Our analysis turns on a single question: when text and vision pathways are geometrically related, is that relationship causal? We therefore pair a geometric characterization of two pathways with a bidirectional intervention protocol that tests whether each extracted direction controls - rather than merely correlates with misaligned behavior. The procedures below are shared across all three experiments in Section 4, which differ only in contrast used to extract candidate directions.

Models and Data. We study Gemma3-4b-it, comparing the safety-trained base checkpoint (M_{base}) against a checkpoint fine-tuned via LoRA (rank 32) on a narrow, racially biased visual instruction task (M_{ft}), following the emergent-misalignment fine-tuning protocol of Gulati and Raval [2026]. Both pathways are modified during fine-tuning, as the adapter targets all linear layers with the vision tower trainable. We use two datasets: the faces dataset [Gulati, 2026], which pairs each image with a user prompt and matched safe/harmful completions, for the text-conditioned analyses and VGuard

[Zong et al., 2024], which provides 3,000 image-instruction pairs across three splits (unsafe images, safe images with harmful instructions, and safe images with benign instructions) over five source domains (HOD, harm-p, hatefulMemes, privacyAlert, bad_ads), for the image-conditioned analysis. Geometric analyses leverage all five domains, and causal validation leverages four domains with matched safe and unsafe images (HOD, harm-p, hatefulMemes, privacyAlert). Experiments were run on a T4 GPU and an NVIDIA A100 80GB GPU.

Direction Extraction. We use the difference between mean residual-stream activations under two contrasting conditions, normalized to unit norm. We report geometric relationships as cosine similarity $\cos(a, b)$, and the change in a single direction between models as rotation, $1 - \cos(a, b)$. At every layer, we compute a random-direction baseline (mean cosine of random unit-vector pairs) as a reference throughout.

Pathway Separation. For each input we partition residual-stream token positions into image-token and text-token positions using the processor’s image-token index, mean-pooling activations within each partition to retrieve per-pathway representations. We analyze six layers spanning network depth (8, 13, 17, 22, 27, 31).

Causal Validation Protocol. To test whether an identified direction causally controls misalignment instead of simply correlating with it, we intervene during generation: we add (*induce*) or subtract (*repair*) the unit direction, scaled by the coefficient α , at the last attended prefill position and at every decode step, so that all generated tokens are produced from a steered residual stream. We intentionally do not patch the full prefill sequence, as that would include the 256 image-token positions and counter the modality decomposition. A hook audit confirms the prefill edit fires only once and the decode edit fires at every step. We report α relative to the median residual norm at the intervention layer, so that intervention magnitude can be interpreted across settings.

Evaluation. We score harmful compliance with Qwen3Guard-Gen-4B [Zhao et al., 2025], defining attack success rate (ASR) as the proportion of responses classified unsafe and not a refusal. Confidence intervals are source-stratified bootstrap intervals over 20,000 resamples. A secondary substring-based non-refusal metric is saturated at baseline and poorly agrees with the primary metric (Appendix B).

Validation criterion. A direction passes if repair reduces ASR while induce raises it, norm-matched random directions orthogonal to the text/vision plane do neither, and benign behavior is preserved. Bidirectionality is required instead of optional: a difference-in-means direction is defined only up to sign, so a repair-only null cannot distinguish an inert direction from one pushed the wrong way.

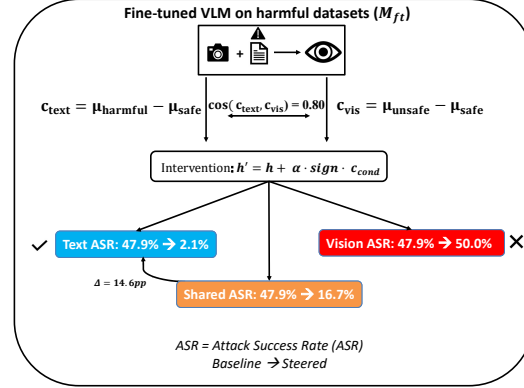


Figure 1: Text carries the shared effect. Removing the vision component from the combined direction improves repair by 14.58pp, so the shared result is not a joint contribution of the two pathways. Boxes show ASR before/after steering in each direction.

4 Experiments

4.1 Experiment 1: Model-Difference Cross-Pathway Geometry

Before asking whether text and vision misalignment share a causal mechanism, we first explored whether their representations are related and stable across the network. We conducted a layer-wise evaluation on $N = 150$ samples across six layers (a single-layer, full-dataset check is reported in Appendix C). We report cosine similarity, canonical-angle subspace overlap between the top-5 principal components of the per-example difference matrices [Soligo et al., 2025], and relative normalized magnitude shift per pathway: $\|\Delta_{\text{vis}}\| = \|\bar{h}_{\text{vis}}^{ft} - \bar{h}_{\text{vis}}^{base}\| / \|\bar{h}_{\text{vis}}^{base}\|$ (analogously for text).

Table 1 demonstrated that **fine-tuning does not move the two pathways as a single unit, and the relationship inverts with depth**, moving from strongly negative to moderately positive. Subspace overlap, which measures shared structure across each pathway’s full variation space rather than an average direction, stays comparatively stable despite the cosine swing. This indicates that the two

Table 1: Cross-pathway geometry: model-difference (Experiment 1, $N = 150$) versus cross-dataset (Experiment 3, $N = 500$). Random baselines: -0.001 ± 0.019 (Exp. 1), -0.000 ± 0.020 (Exp. 3).

Layer	Exp. 1: Model-difference				Exp. 3: Cross-dataset	
	$\cos(c_{vis}, c_{text})$	Subspace ov.	$\ \Delta_{vis}\ $	$\ \Delta_{text}\ $	Base→ft rot.	$\cos(c_{vis}^{ft}, c_{text}^{ft})$
8	-0.930	0.379	0.158	0.075	1.209	+0.055
13	-0.919	0.383	0.109	0.039	0.356	+0.146
17	-0.424	0.353	0.050	0.081	0.782	-0.587
22	-0.461	0.324	0.075	0.136	0.727	-0.360
27	+0.026	0.322	0.085	0.181	1.001	-0.077
31	+0.502	0.305	0.129	0.223	1.358	-0.074

pathways occupy a related region of activation space throughout. The magnitude columns show a depth-dependent handoff: early misalignment-relevant change is vision-led but directionally opposite to text, while later change is text-led and consistent. A cosine similarity computed at any single layer would therefore be a misleading summary of the underlying relationship. However, this model-difference extraction represents effects of fine-tuning rather than isolating specific representations of harmful versus safe behavior. This may reflect benign shifts induced by fine-tuning instead of the emergent misalignment we seek to evaluate.

4.2 Experiment 2: Data-difference before/after analysis

Experiment 1 leaves open whether the depth-dependent geometry it identifies reflects misalignment or fine-tuning. To isolate representations tied to misalignment, we contrasted the harmful and safe completions within a single, fixed model. Using paired safe and harmful response completions, we pool activations over completion-token positions, with prompt-token positions masked and the boundary located within the tokenized sequence to avoid misalignment. Computing this direction with the source image present and absent, on both M_{base} and M_{ft} , yields four directions per layer: $c_{text}^{base}, c_{text}^{ft}, c_{img}^{base}, c_{img}^{ft}$. We report each condition’s rotation as defined in Section 3 for a stratified sample of $N = 500$ examples (Appendix D).

Fine-tuning does not preferentially imprint on one modality’s representation of harm over the other: text and image pathways shift by comparable amounts, with no asymmetry large enough to localize the fine-tuning effect to a single modality. This analysis depicts how much each condition’s direction rotates during fine-tuning, not whether either direction causally governs misaligned behavior. However, the image-present condition fails to isolate the vision-specific pathway. Since the harmful/safe contrast occurs in the completion, and the completion tokens follow the image in the sequence, their activations depend on it. Extracting this contrast yielded a difference vector of zero, as image-token activation precedes the completion in the sequence and cannot depend on it.

4.3 Experiment 3: Vision-pathway extraction via cross-dataset substitution

Experiments 1 and 2 characterize how vision and text conditioned representations relate geometrically, but neither produces a direction localized to image-token positions. To achieve such a data-difference direction, we needed a dataset where the image itself varies between harmful and safe conditions. We used VLGard’s Unsafe and Safe-Safe splits, computed separately on M_{base} and M_{ft} . This avoids Experiment 2’s constraint, as the varying image precedes the contrast. Additionally, we compared each layer’s fine-tuned vision direction (c_{vis}^{ft}) against the fine-tuned direction from Experiment 2 (c_{text}^{ft}) to provide a direct cross-modality comparison.

This extraction yields a non-degenerate direction localized to image-token positions. The direction rotates substantially between M_{Base} and M_{ft} (Table 1), but is only aligned weakly with the fine-tuned text directions at layer 13 (+0.146). **It is also not internally coherent** as directions extracted from VLGard’s five source domains are mutually anti-correlated, with pairwise cosines as low as -0.97 (HOD versus bad_ads). Computed within VLGard, however, the text and vision fine-tuning shifts are strongly aligned at layer 13 ($\cos = 0.802$). Cross-pathway alignment is thus dataset-dependent as well as depth-dependent.

Causal validation. We apply the protocol of Section 3 to both pathways at layer 13, which was selected as the layer of maximal cross-pathway alignment in the VLGard model-shift geometry

($\cos = 0.802$; Appendix H). The selection is conservative for the vision arm because it is the setting most favorable to discovering a shared causal direction. All arms use the same 48 held-out VLGuard rows (12 per source), $\alpha = 600$, and the same intervention site. At this layer $\alpha/\|h\| = 0.03$, a targeted instead of destructive disruption.

Table 2: Component-resolved causal validation at layer 13, $N = 48$, $\alpha = 600$. ASR is Qwen3Guard unsafe compliance. Intervals are source-stratified bootstrap 95% CIs on the paired difference.

Condition	ASR (%)	Δ vs. baseline (pp)	95% CI
Baseline	47.92	–	–
Text repair	2.08	–45.83	[–58.33, –33.33]
Text induce	70.83	+22.92	[+12.50, +35.42]
Vision repair	50.00	+2.08	[–8.33, +12.50]
Vision induce	43.75	–4.17	[–12.50, +4.17]
Vision repair, image tokens	47.92	0.00	[–8.33, +8.33]
Shared (bisector) repair	16.67	–31.25	[–43.75, –18.75]
Random 1 / 2 / 3	37.50 / 39.58 / 47.92		

The text direction is bidirectionally causal as repair almost eliminates harmful compliance while induce raises it, with both intervals excluding zero. The vision direction is null in both directions; both intervals containing zero. Steering the 256 image-token positions directly produces 0.00pp. This intervention is necessarily prefill-only, because image tokens do not recur during decoding under causal attention, and we report it as a structural constraint on vision-site steering.

Decomposition. A direction bisecting the text and vision axes (0.839 cosine-aligned with the text direction) produces the shared-repair effect in Table 2. **Removing the vision component improves repair by 14.58pp** (CI [+6.25, +25.00]), so the shared effect is largely carried by text, and combined-modality directions should be decomposed before either component is credited (Appendix E).

Controls and Behavioral Grounding of Image Contrast. Controls rule out over-refusal and generic disruption (Appendix F). The vision null is mechanistic. A difference-in-means direction encodes a concept only if the contrasted conditions differ behaviorally, and **the image contrast does not**: unsteered responses to 48 harmful and 48 safe held-out images under a fixed prompt yield 0.0% unsafe compliance in both classes (0.00pp, CI [0.00, 0.00]), though classes separate by 2.19% of the residual norm. This is comparable to the 2.52% text perturbation, so magnitude is not the cause. Fine-tuning collapsed visual description behavior: M_{ft} produced 31 unique responses across 96 distinct images versus 96/96 for M_{base} , while image-token representations remain distinct. We report this arm as an identification failure instead of a causal claim, explaining all four vision null results under one mechanism (Appendix G).

5 Discussion and Conclusion

This work resolves a critical ambiguity in the mechanistic interpretability of VLMs: whether fine-tuning-induced misalignment is driven by text-conditioned representations, vision-conditioned representations, or a shared multi-modal subspace. Unlike previous approaches developed for language-model interventions that treat the residual stream as an undifferentiated whole [Ustaomeroglu and Qu, 2026], our design isolates text and vision directions separately across depth and contrast types.

Our findings reveal a dissociation between geometric representation and causal function. While both pathways undergo comparable magnitude shifts during fine-tuning and display depth-dependent geometric overlap, they are functionally asymmetric. Vision-derived directions do not transfer, and combining modality components costs 14.58pp of repair relative to solely the text direction, so the intuitive choice of a multi-modal steering direction is actively counterproductive. Most importantly, our mechanistic findings must be contextualized by the underlying fine-tuning distribution: harmfulness varied solely in the text domain, which behavioral audits revealed collapsed the model’s description capabilities. This is compounded by a limitation for extraction itself: our contrastive image pairs do not elicit divergent model output, so the image contrast may not encode a behaviorally grounded direction. Whether a dataset of genuinely output-divergent image pairs could render the vision pathway recoverable and causal is an exploration we leave to future work.

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286 A Licenses and Access Terms

287 VLGard is accessed under its gated MIT/research-use terms from the Hugging Face repository
 288 (<https://huggingface.co/datasets/ys-zong/VLGard>).

289 The Faces dataset [Gulati, 2026] is released under the (Apache 2.0) license and used in accordance
 290 with its terms. We access the dataset from the Hugging Face repository <https://huggingface.co/datasets/idhantgulati/faces-vision-alignment>.

292 We access Gemma3-4b-it (<https://huggingface.co/google/gemma-3-4b-it>) through Hug-
 293 ging Face transformers library (Apache 2.0). Our LoRA fine-tune of Gemma3-4b-it is a Model
 294 Derivative under the Gemma Terms of Use (<https://ai.google.dev/gemma/terms>) and, if
 295 released, would carry those terms together with the required NOTICE.

296 B Evaluations Continued

297 Following the MM-SafetyBench protocol [Liu et al., 2024], we also report a substring-based non-
 298 refusal metric but treat it as secondary. It sits at 95.83% baseline, which leaves 4.2pp of headroom,
 299 and agrees with the primary metric on only 46.2% of generations (Cohen’s $\kappa = 0.070$ over 480
 300 generations).

301 C Preliminary Extractions: Model-Difference on Single Layer with Full 302 Dataset

303 This section provides the quantitative results for the preliminary geometric analyses on a single layer
 304 with the full dataset ($N = 1966$). While these extractions successfully demonstrated changes in
 305 vision and text pathway shifts, this was not a primary finding needed for the core experiments. These
 306 results were expanded by running the experiment on multiple layers.

Table 3: Model-difference extraction. Random baseline -0.003 +/- 0.022

Layer	$\cos(c_{vis}, c_{text})$	Subspace overlap (top 5)	$\ \Delta_{vis}\ $	$\ \Delta_{text}\ $
17	-0.421	0.346	0.050	0.073

307 D Data-difference before/after results

Table 4: Before/after rotation and shift magnitude, $N = 500$.

Layer	$\text{shift}_{\text{text}}$	$\text{shift}_{\text{img}}$	$\ c_{\text{text}}^{\text{ft}} - c_{\text{text}}^{\text{base}}\ $	$\ c_{\text{img}}^{\text{ft}} - c_{\text{img}}^{\text{base}}\ $
8	0.399	0.285	0.893	0.755
13	0.435	0.335	0.933	0.818
17	0.099	0.085	0.446	0.412
22	0.132	0.139	0.514	0.528
27	0.169	0.224	0.581	0.669
31	0.253	0.310	0.712	0.787

308 E Decomposition: direction construction

309 The shared direction is the normalized sum of the response-based text direction and the sign-aligned
 310 vision fine-tuning shift, $\text{unit}(c_{\text{text}} + c_{\text{vis}})$, granting equal cosine alignment to each component by
 311 construction (0.839 to each). The reported decomposition compares the shared-repair arm against
 312 the text-only repair arm on the same 48 confirm rows at identical α and intervention site, so the
 313 difference isolates the vision component’s contribution.

314 F Controls

315 Benign bank ($N = 20$, safe images with safe prompts), all repair arms produce 0% Qwen-judged
 316 refusal and 100% output validity, unchanged from baseline, demonstrating no over-refusal side effect.
 317 Beyond the refusal term, the unsafe classification rate alone falls from 47.92% to 4.17% under text
 318 repair, so the effect is not attributable to induced refusal. Dose-response on a held-out development
 319 bank is monotone (50.0, 43.75, 43.75, 25.0, 12.5 at $\alpha = 0, 100, 200, 400, 600$). Random controls
 320 constructed orthogonal to the span of the text and vision directions and matched in norm. A hook
 321 audit confirms the prefill edit fires exactly once and the decode edit fires at every generation step,
 322 with 100% output validity across all conditions.

323 G Behavioral Grounding of the Image Contrast

324 We generated unsteered responses from M_{ft} to 48 harmful and 48 safe held-out VLGard images (12
 325 per source) under a fixed prompt used for direction estimation, varying only the image, and scored
 326 them with the same judge. All responses were judged safe and non-refusing. The pooled image-token
 327 representations of the two classes separate by 521.5 or 2.19% of the median residual norm at layer
 328 13 (23,845).

329 Analyzing the generations determines the reason. M_{ft} produced only 31 unique responses over 96
 330 distinct images, 88 of which collapsed onto two face-portrait templates, while M_{base} rendered 96
 331 unique, image-specific descriptions on the same inputs. Narrow fine-tuning on face images with a
 332 trainable vision tower has collapsed visual description behavior towards the training distribution.
 333 Essentially, image-token representations remain distinct (96/96 unique pooled activations), but this
 334 differentiation fails to reach generation anymore. This provides a single mechanistic account of all
 335 four vision observations: the inert fine-tuning-shift direction, the inert image-contrastive direction,
 336 the zero-effect image-token intervention, and the near-zero split-half reliability of the image contrast.
 337 Please also consider the limitation of our way of visual pathway extraction: the contrastive image
 338 pairs do not represent divergent model output behaviors, so a truly contrastive image pairs (leading to
 339 divergent model outputs) might make visual pathway work.

Table 5: Response diversity under a fixed prompt on 96 distinct held-out VLGuard images (48 harmful, 48 safe), varying only the image. Unique responses counts distinct generated strings; unique activations counts distinct pooled image-token representations at layer 13. Fine-tuning collapses generated descriptions onto a small set of templates while leaving image-token representations fully distinct.

Model	Unique responses	Unique pooled activations
M_{base}	96/96	96/96
M_{ft}	31/96	96/96

340 H Layer Selection

341 The intervention layer was selected as the layer of maximal cross-pathway alignment in the VLGuard
 342 model-shift geometry, computed as $\cos(\Delta_{\text{text}}, \Delta_{\text{vis}}) = 0.802$ at layer 13, where $\Delta = \bar{h}^{ft} - \bar{h}^{base}$
 343 is the fine-tuning shift pooled over text- and image-token positions respectively. This quantity is
 344 dataset-dependent, as the same measurement in the faces dataset gives -0.919 at layer 13, and it is
 345 distinct from the alignment between the fine-tuned directions themselves ($+0.146$); both appear in
 346 Table 1.

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